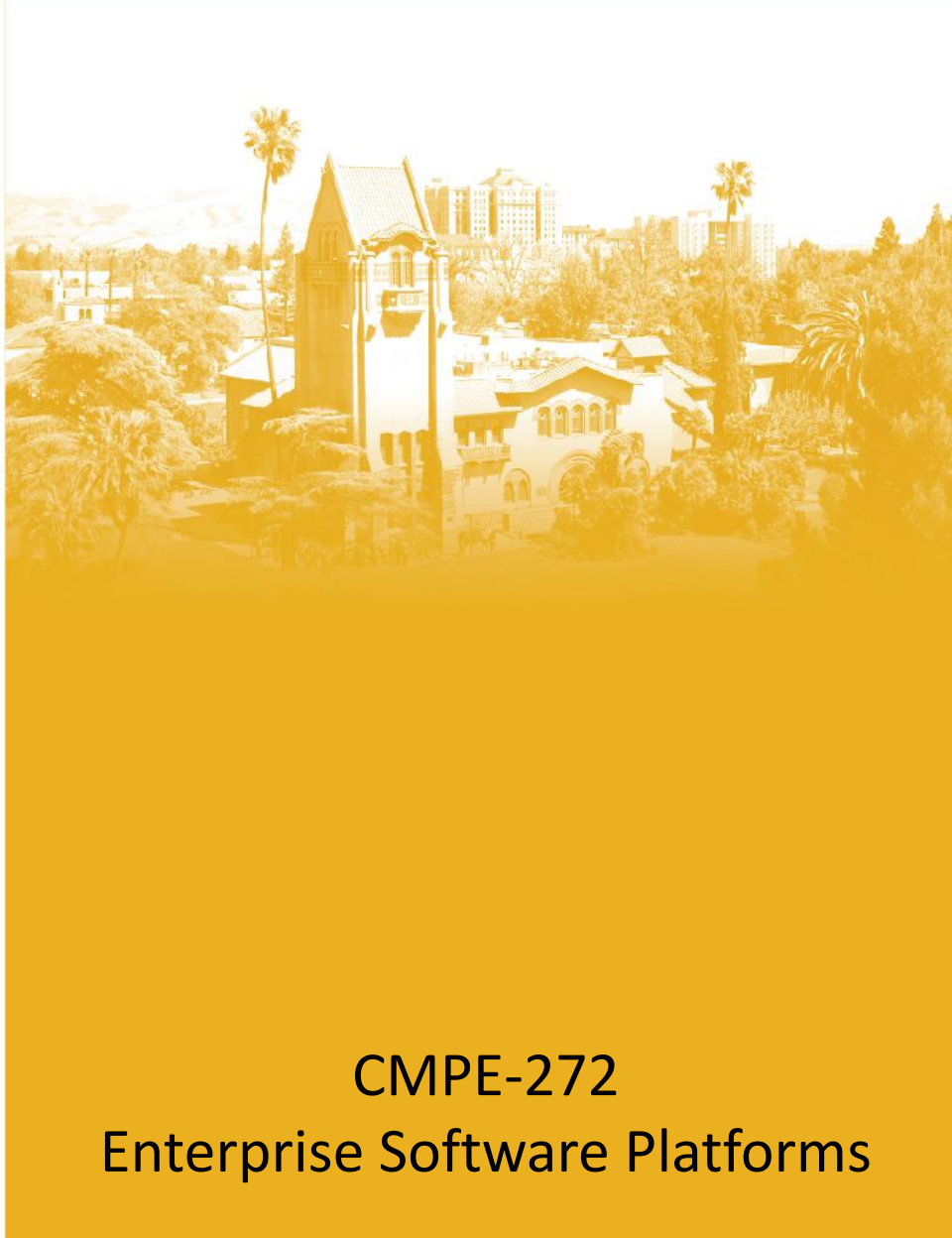




CMPE-272

Enterprise Software Platforms

Enterprise Software Platforms

Instructor: Andrew Bond

IT News

- Dreamforce 2026, San Francisco, September 15 to 17. Salesforce presented the CRM as a place where AI agents do the work, not only a record of customers.
 - Alforce: a layer that lets outside AI agents act inside Salesforce, through an API, MCP or a CLI, under Salesforce's own permissions.
 - MCP security: integrations are scanned when an agent registers them and rated for prompt injection and tool poisoning.
- SAP, September 2: HARTING moves to the cloud with RISE with SAP. Mainstream maintenance for SAP ECC ends in 2027, so migrations like this one fill SAP's news this year.
- SAP, September 15: TabPFN-3.5 Plus, a prediction model for tabular data, is now available in SAP AI Core, so predictions run next to the ERP data.
- Tonight's question: when the ERP or CRM vendor ships the AI agent, who owns the business process, your team or the vendor?

UNIT 5 · ENTERPRISE APPLICATIONS: ERP, SCM AND CRM (WEEK 6)

The Big Three Today

- **SAP S/4HANA:** in-memory ERP; RISE with SAP; ECC end-of-maintenance (2027) is forcing migrations now
- **Salesforce:** CRM grown into a platform — Data Cloud, Agentforce, the AppExchange ecosystem
- **Microsoft Dynamics 365 + Power Platform:** ERP/CRM woven into M365 and Azure
- Also running the world: **Oracle Fusion / NetSuite, Workday** for HR and finance

Extending Without Forking: the Clean Core

- Customization debt is why ERP upgrades take years — modifications welded into the core
- Clean core: keep the vendor core vanilla; extend via platform layers (SAP BTP, Salesforce Platform, Power Platform)
- Extension points are the patterns you already know: events, APIs, side-by-side apps (Units 2 and 6)

Why Replacing an ERP Is So Hard

- **Data gravity:** decades of transactions, master data, and process assumptions
- **The processes ARE the org chart:** Change the system and you change how people work
- **Big-bang vs phased cutover:** Hershey and Lidl as canonical failure case studies
- **Realistic posture:** strangler-fig around the edges; rip-and-replace is rarely the answer

INTRODUCTION TO ENTERPRISE SYSTEMS

Functional Structure

- The most common enterprise structure you are likely to encounter is the Functional Structure
- Organizations that utilize a functional structure are divided into functions, or departments, each of which is responsible for a set of closely related activities
- E.g., the accounting department sends and receives payments, and the warehouse receives and ships materials.
- Typical functions or departments found in a modern organization include
 - *Purchasing*
 - *Operations*
 - *Warehouse*
 - *Sales and marketing*
 - *research and development*
 - *finance and accounting*
 - *human resources*
 - *information systems*



Silo Effect

- People in the different functional areas came to perform their steps in the process in isolation, without fully understanding which steps happen before and which steps happen next
- They essentially complete their part of the process, hand it off to the next person, and then proceed to the next task
- By focusing so narrowly on their specific tasks, they lose sight of the “big picture” of the larger process, be it procurement, fulfillment, or any number of other common business processes
- This tendency is commonly referred to as the **silo effect** because workers complete their tasks in their functional “silos” without regard to the consequences for the other components in the process
- It becomes a major challenge for enterprises to coordinate activities among the different functional areas

Enterprise Systems

- Business processes span different parts of an organization
- Various process steps are increasingly executed by groups in multiple locations throughout the world
- E.g., a bicycle manufacturer may purchase components from Italy, produce bicycles in Germany, and sell those bicycles in the United States
- Steps in business processes are performed in locations that are geographically dispersed, making it impossible to manage such processes effectively without the use of modern information systems
- Systems that support end-to-end processes are called **Enterprise Systems (ES)**, and they are essential to the efficient and effective execution and management of business process

Augmenting Human Intellect (Groupware)

- By "augmenting human intellect" we mean increasing the capability of a man to approach a complex problem situation, to gain comprehension to suit his particular needs, and to derive solutions to problems.
- Application software designed to help people involved in a common task to achieve their goals

<http://www.douengelbart.org/pubs/augment-3906.html>

<http://nexus.awakentech.com:8080/at/awaken1.nsf/UNIDs/CFB70C1957A686E98825654000699E1B?OpenDocument>

The Original Groupware

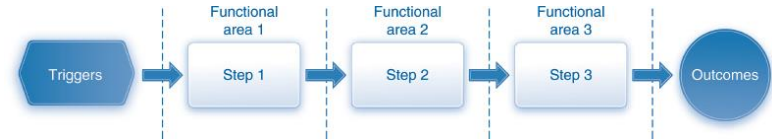


- **IBM Notes** (formerly **Lotus Notes**; see [branding](#), below) and **IBM Domino** (formerly **Lotus Domino**^[1]) are the [client](#) and [server](#), respectively, of a [collaborative client-server software platform](#) marketed by IBM.
- [Lotus Notes 8 Demo](#)

ENTERPRISE SYSTEMS

- Business Processes

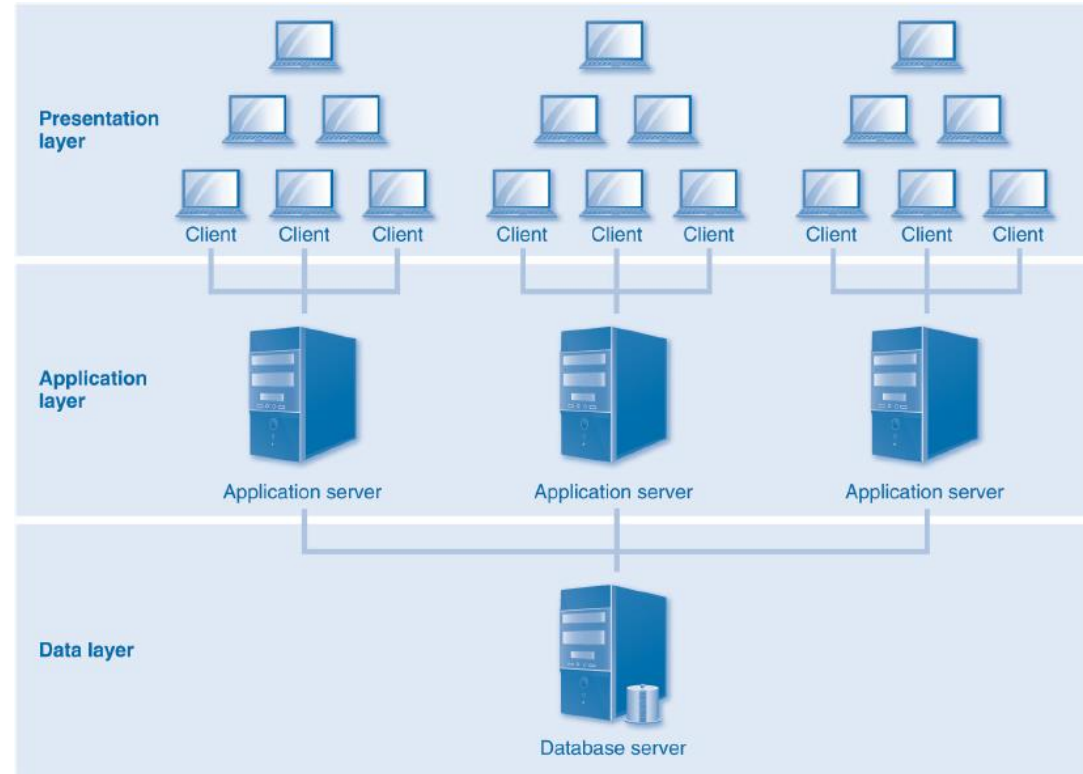
- **Procurement process** (*buy*)
- **Production process** (*make*)
- **Fulfillment process** (*sell*)
- **Lifecycle data management process** (*design*)
- **Material planning process** (*plan*)
- **Inventory and warehouse management (IWM) process** (*store*)
- **Asset management and customer service processes** (*service*)
- **Human capital management (HCM) processes** (*people*)
- **Project management processes** (*projects*)
- **Financial accounting (FI) processes** (*track–external*)
- **Management accounting or controlling (CO) processes** (*track–internal*)



Architectures: Client-Server

Layers:

- (1) how you interact with the application (Presentation)
- (2) what the application allows you to do (Application);
- (3) where the application stores your work (Data)

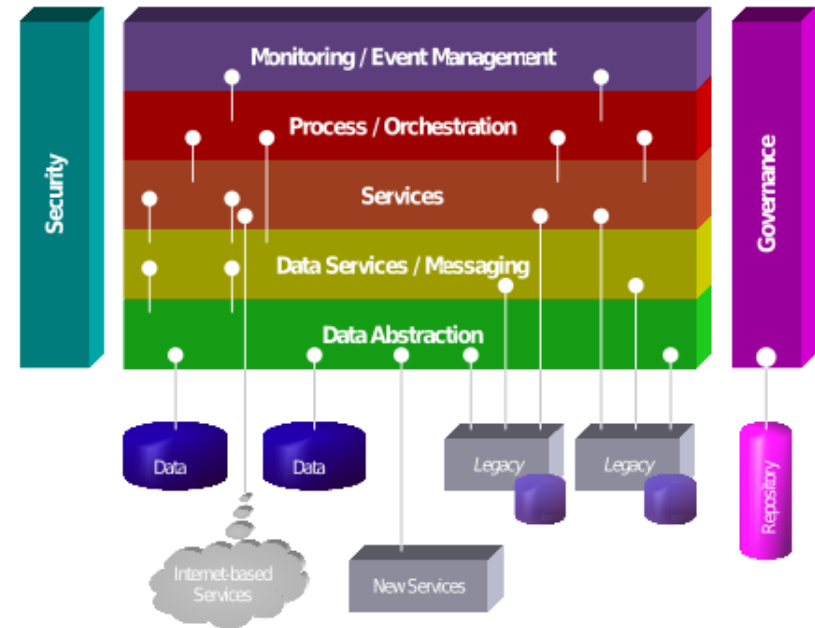


Architectures: Service-Oriented Architecture (SOA)

- Extension of Client-Server Architecture
- Integrate multiple client-server applications and create enterprise *mash-ups*, or *composite applications*

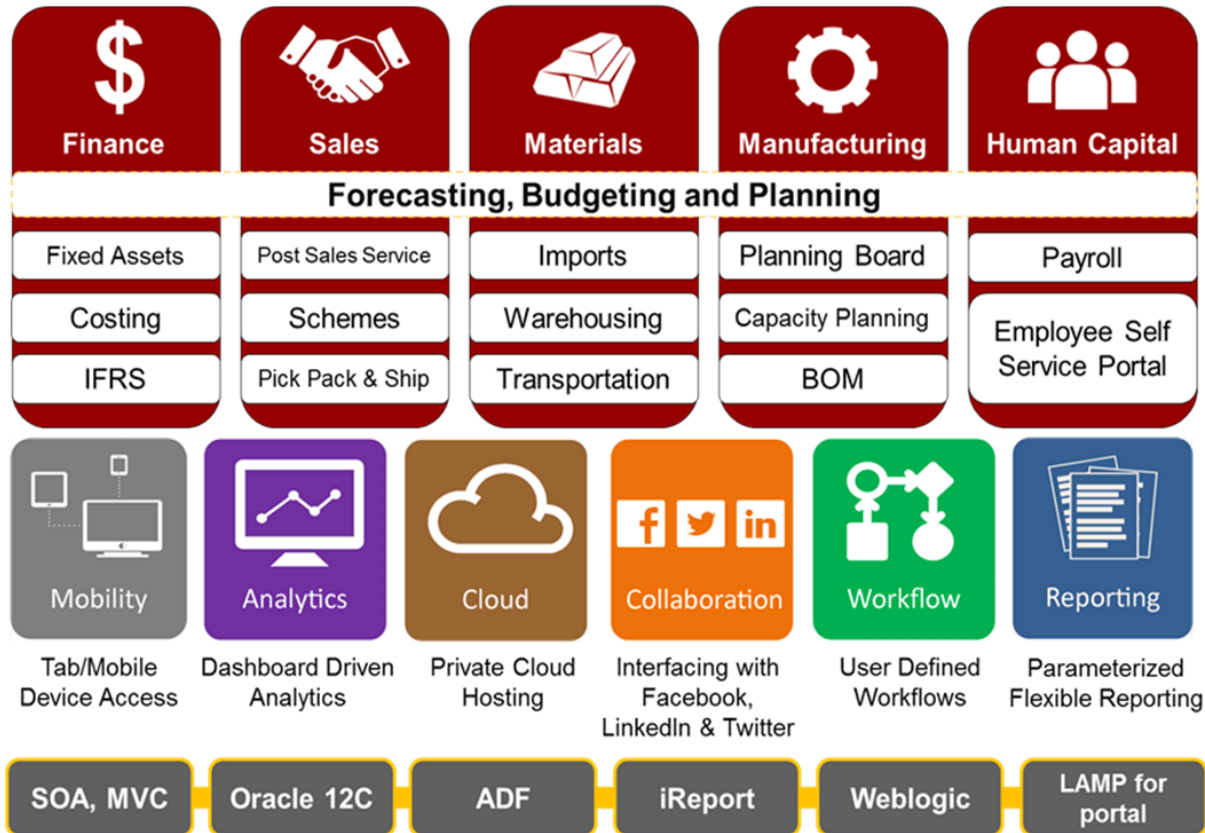
Four properties of a Service

1. It logically represents a business activity with a specified outcome.
2. It is self-contained.
3. It is a [black box](#) for its consumers.
4. It may consist of other underlying services. [\[3\]](#)



Architectures: ERP

- ERP systems focus primarily on intra-company processes—that is, the operations that are performed within an organization—and they integrate functional and cross-functional business processes



ENTERPRISE RESOURCE PLANNING -ERP

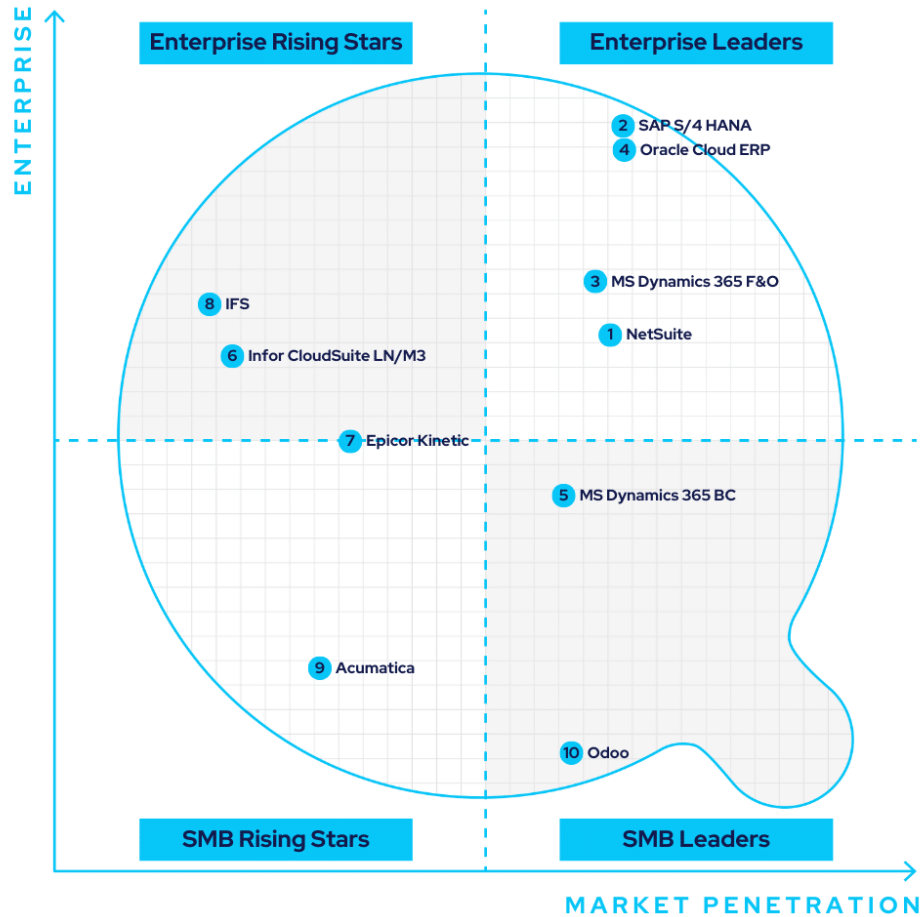
What is ERP?

- The practice of consolidating an enterprise's planning, manufacturing, sales and marketing efforts into one management system.¹
- Combines all databases across departments into a single database that can be accessed by all employees.²
- ERP automates the tasks involved in performing a business process.¹

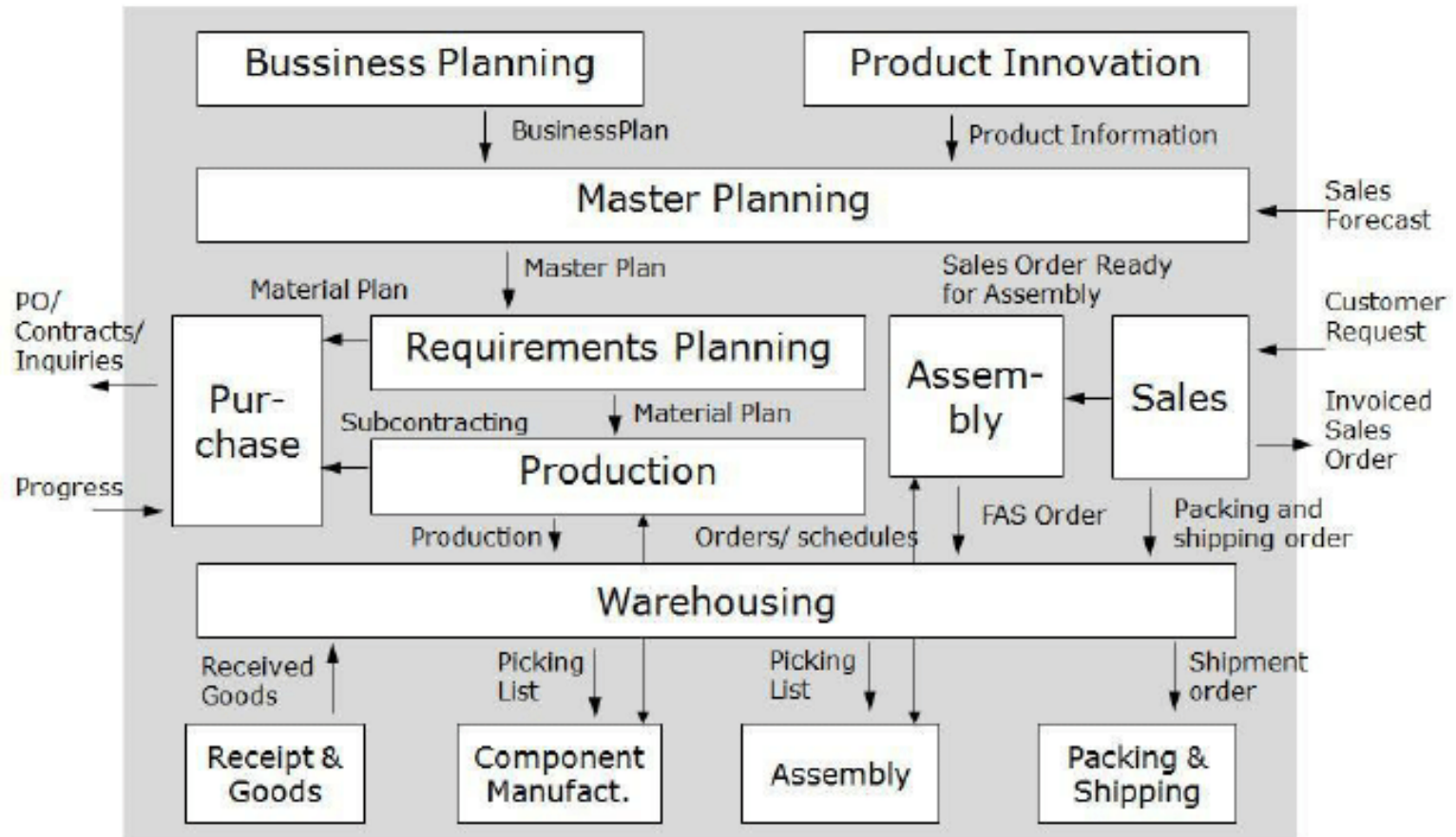
Sources:

1. <http://www.cio.com/summaries/enterprise/erp/index.html>
2. CIO Enterprise Magazine, May 15, 1999.

Top 10 ERP Systems in 2025



How Do ERP Systems Work?



ERP Components

- Finance: modules for bookkeeping and making sure the bills are paid on time. Examples:
 - General ledger
 - Accounts receivable
 - Accounts payable
- HR: software for handling personnel-related tasks for corporate managers and individual employees. Examples:
 - HR administration
 - Payroll
 - Self-service HR
- Manufacturing and Logistics: A group of applications for planning production, taking orders and delivering products to the customer. Examples:
 - Production planning
 - Materials management
 - Order entry and processing
 - Warehouse management

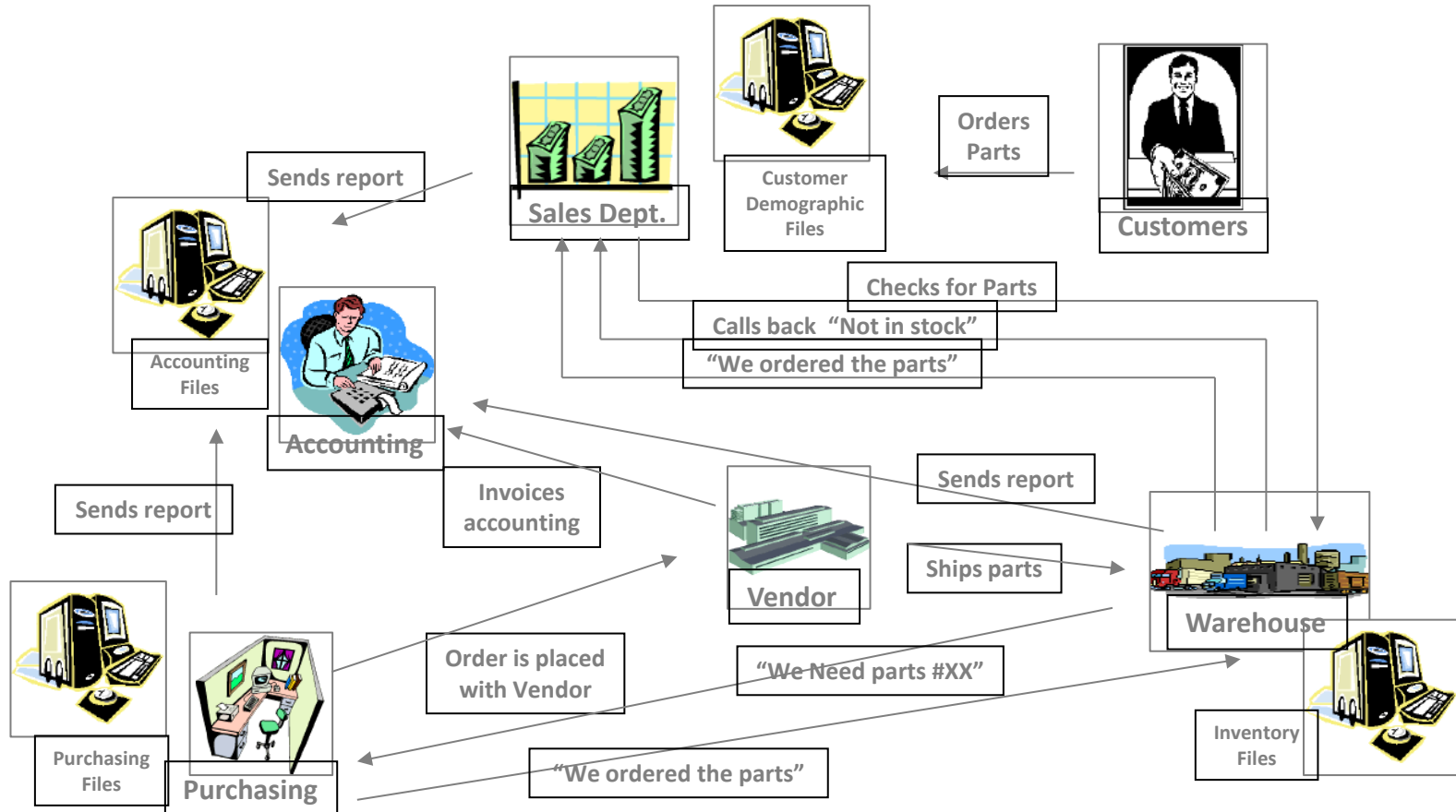
Modules of ERP systems

- **Enterprise Resource Planning (ERP) Systems:**
 - Software packages that can be used for the core systems necessary to support enterprise systems.
 - SCM
 - CRM
 - HRM
 - Collaboration
 - Content Management
 - Business Intelligence
 - Identity Management

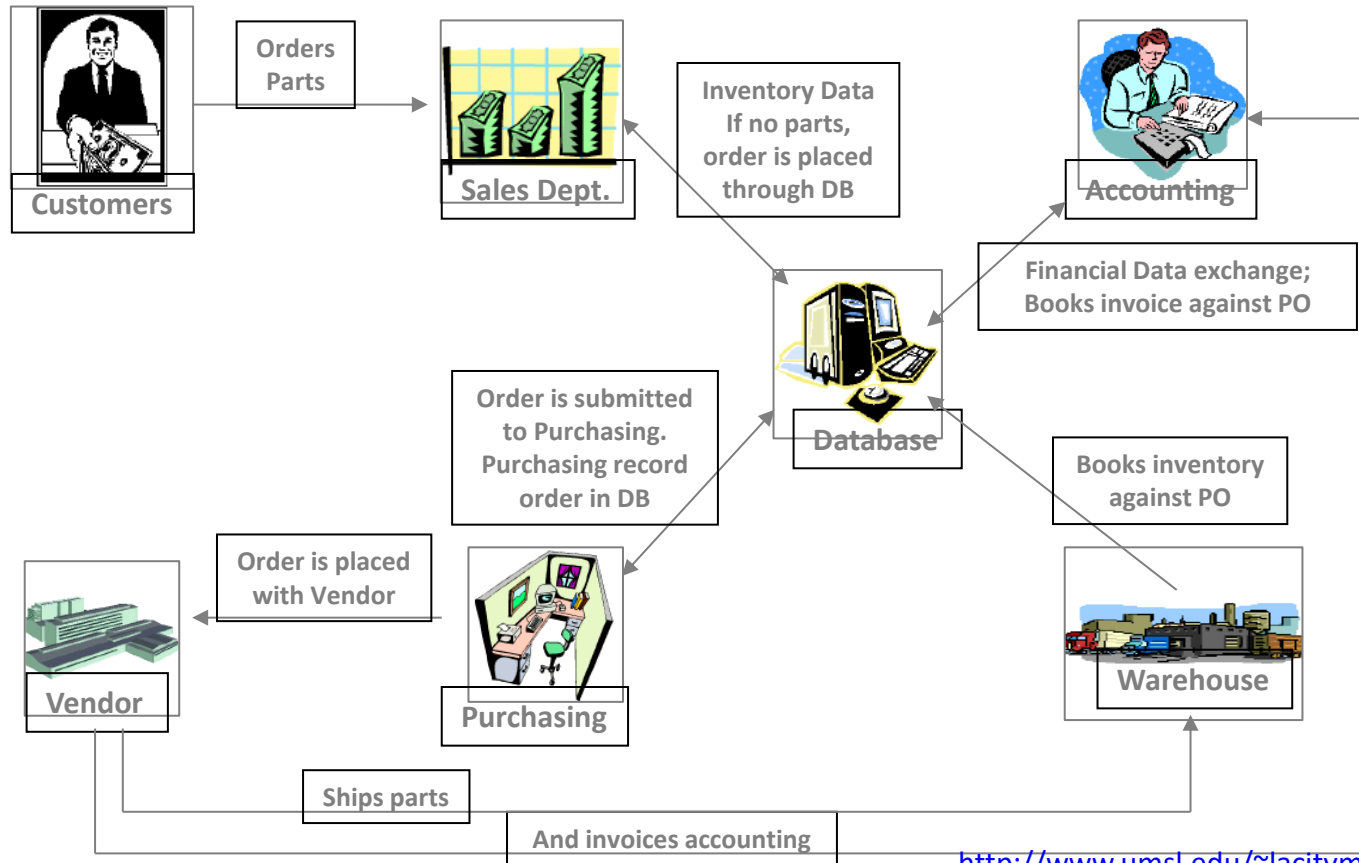
ENTERPRISE RESOURCE PLANNING



An ERP Example: Before ERP



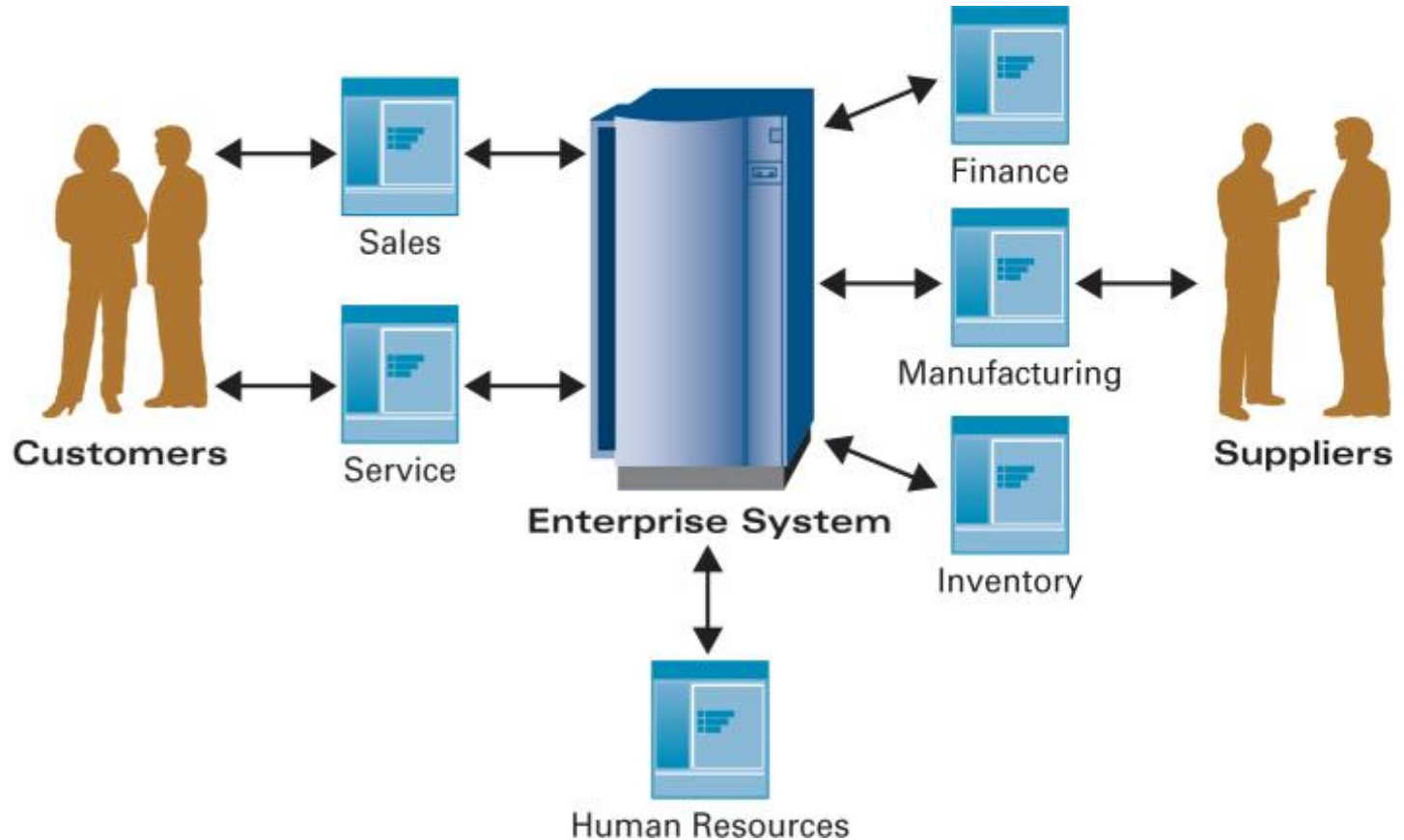
An ERP Example: After ERP



ERP

- Attempts to integrate everything
 - CRM drives what SCM will produce
 - Everyone works together in e-collaboration
 - The entire organization knows the entire organization
- Think about your school
 - Can you register for class with a bill outstanding?
 - Can you register for a class for which you haven't completed the prerequisite?

ERP Integrates Everything



SD Audit Trail for Completion of Steps in the SAP Sales Process

Document	On	Status
▼  Quotation 0020000019	10/16/2007	Completed
▼  ➔ Standard Order 0000012071	10/22/2007	Completed
▼  Delivery 0080015185 (Schedule delivery)	10/22/2007	Completed
 WMS transfer order 0000001511 (Picking ticket)	10/22/2007	Completed
 GD goods issue:delvy 4900035532 (Shipment)	10/22/2007	complete
▼  Invoice (F2) 0090036264	10/23/2007	
 Accounting document 1400000000	10/23/2007	Cleared

Audit Trail for Completion of Steps in the SAP Purchase Process

Standard PO 4500017175 Vendor 62999 GPSRUS-999 Doc. date 11/06/2007

Header
Item Overview
Item [10] TG73602999, GPS Guidance System-999

Material Data Quantities/Weights Delivery Schedule Delivery Invoice Conditions Purchase Order History Texts

Sh...	MvT	Material Do...	Item	Posting Date	Quantity	Delivery cost quantity	OU	Amount in LC	L.cur	Qty in OPUn
WE	101	5000011991	1	11/06/2007	15	0	EA	40,500.00	USD	15
Tr./Ev. Goods receipt					15		EA	40,500.00	USD	15
RE-L		5105808722	1	11/12/2007	15	0	EA	40,500.00	USD	15
Tr./Ev. Invoice receipt					15		EA	40,500.00	USD	15

(Purchase order)

(Vendor invoice)

(Goods receipt)

Pros of Enterprise Systems

- Single database
- Integrated system (e.g., capability to determine ATP)
- Process orientation (versus function orientation)
- Standardization of business processes and data, easier to understand across the organization
- Faster business processes (e.g., customer fulfillment, product development)
- Timely information
- Better financial management (partly due to integration)
- One face to the customer
- Optimal inventory levels
- Improved cash management
- Productivity improvements, reduced personnel expense
- Improved financial disclosures
- Improved budgeting, forecasting, and decision support
- Seamless integration and accessibility of information across the organization
- Catalyst for reengineering old, inefficient business processes

Pros of ERP Packages

- One package across many functions (if one ERP)
- "Best practices"
- Modular structure (buy what you need)
- No development needed (unless modifications are required)
- Configurable
- Reduced errors (e.g., business rules, enter data once)

Cons of Enterprise Systems and ERP Packages

Cons of Enterprise Systems

- Centralized control versus decentralized empowerment
- Inability to support traditional business processes that may be best practices for that organization
- Loss of flexibility in rapidly adapting to desired new business processes in the post-implementation period
- Increased complexity of maintaining security, control, and access permissions for specific information embedded in central database
- The rigidity of "standardization" can impede creative thinking related to ongoing business process improvements

Cons of ERP Packages

- Complex and inflexible
- Best practices are shared by all who buy
- Difficult to configure
- Long implementation
- Best of breed might be better (than single ERP package)
- Can't meet all needs (i.e., developed for many user types)

CRM

Customer Relationship Management (CRM) Systems

- Customer relationship management (CRM) systems, sometimes called e-CRM systems, use technology to help an e-business manage its customer base
- CRM allows an e-business to match customer needs with product plans and offerings, remind customers of service requirements, and determine what products a customer has purchased

CRM and Marketing Management

► **Marketing resource and marketing management**

- Manage and optimize the use of marketing resources including budgets, people, time, and assets.
- Align all activities and resources around strategic marketing goals.
- Gain visibility and control into your marketing processes.
- Accurately manage the marketing budgets and costs.
- Increase brand awareness with proper usage and consistency across enterprise and third-party agencies.
- Facilitate collaboration among team members and coordinate marketing activities across the enterprise.

Data Elements in SAP ERP

Today we will discuss

- How data is managed in SAP ERP?
- Understanding the components of creating a sales order in ERP.
- Understanding the steps involved in creating the standard order in SAP ERP.
- The overall order management process in ERP.
- Importance of CRM in ERP Systems.

Definitions

- **SAP S/4HANA** — “SAP Business Suite 4 SAP HANA,” SAP’s current ERP built on the in-memory HANA database (High-Performance Analytic Appliance)
BP — Business Partner (the unified master record for customers/vendors)
MM — Materials Management
PP — Production Planning
SD — Sales and Distribution
FI — Financial Accounting
AR — Accounts Receivable (often written FI/AR when referring to FI’s A/R subledger)
AP — Accounts Payable
WM — Warehouse Management
EWM — Extended Warehouse Management
MRP — Material Requirements Planning
SKU — Stock Keeping Unit (a product identifier)
UoM — Unit of Measure
GR/IR — Goods Receipt/Invoice Receipt (clearing account that reconciles received goods with supplier invoices)
VAT — Value-Added Tax
MDG — Master Data Governance (workflow/governance for master data)
Price control “S/V” — **S** = Standard price; **V** = Moving-average price

Master Data

Long-term data represent entities associated with various processes

- Customer
- Vendor
- Material

Centrally stored and is processed to eliminate data redundancy.

Typically includes

- General Data (across company's code)
- Financial Data
- Area-Specific Data

Transactional Data

Data generated during execution of a process is termed as transactional data.

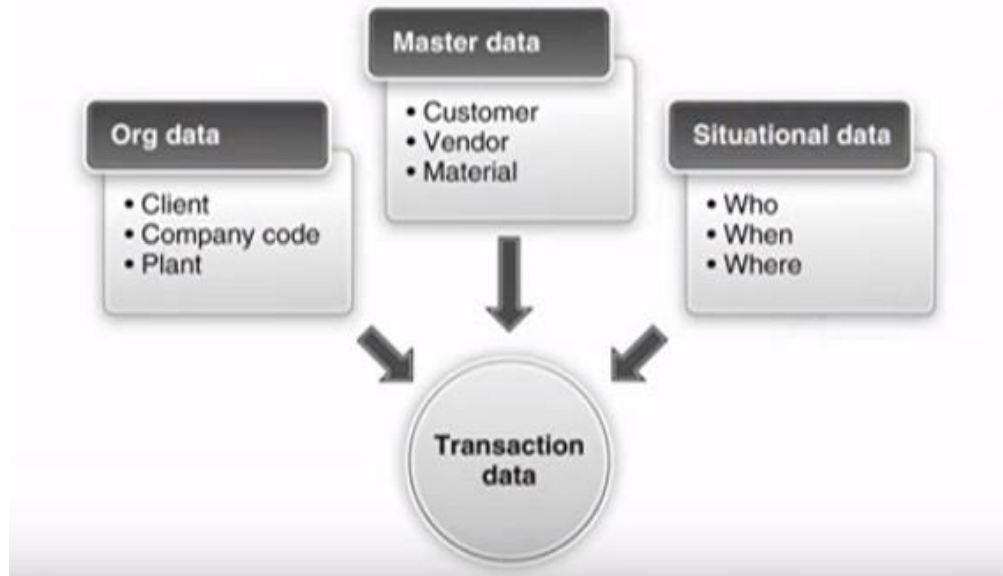
It requires

- Organizational Data
- Master Data
- Situational Data (What, when and how is it happening?)

Example: Sales Order Creation

- Organizational elements:
 - Client,
 - Company Code
 - Sales Area
- Master Data
 - Customer
 - Material
- Situational Data (What, when and how is it happening?)
 - Date
 - Time
 - Person

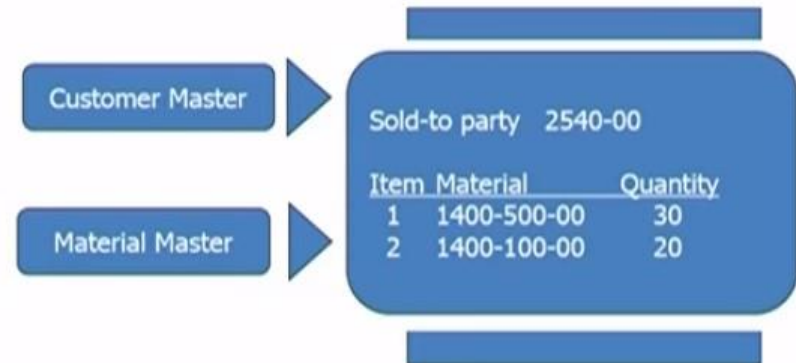
Example of SAP ERP Data



Creating Order in ERP

The three types of master data critical to sales order processing

1. Customer – Shared with Financial Module
2. Material – Shared with Material Management and Production Planning
3. Pricing



Creating Order in ERP

When sales order is created

1. Customer Master
2. Material Master
3. Condition records for Pricing
4. Tables in Configuration for shipping point and route

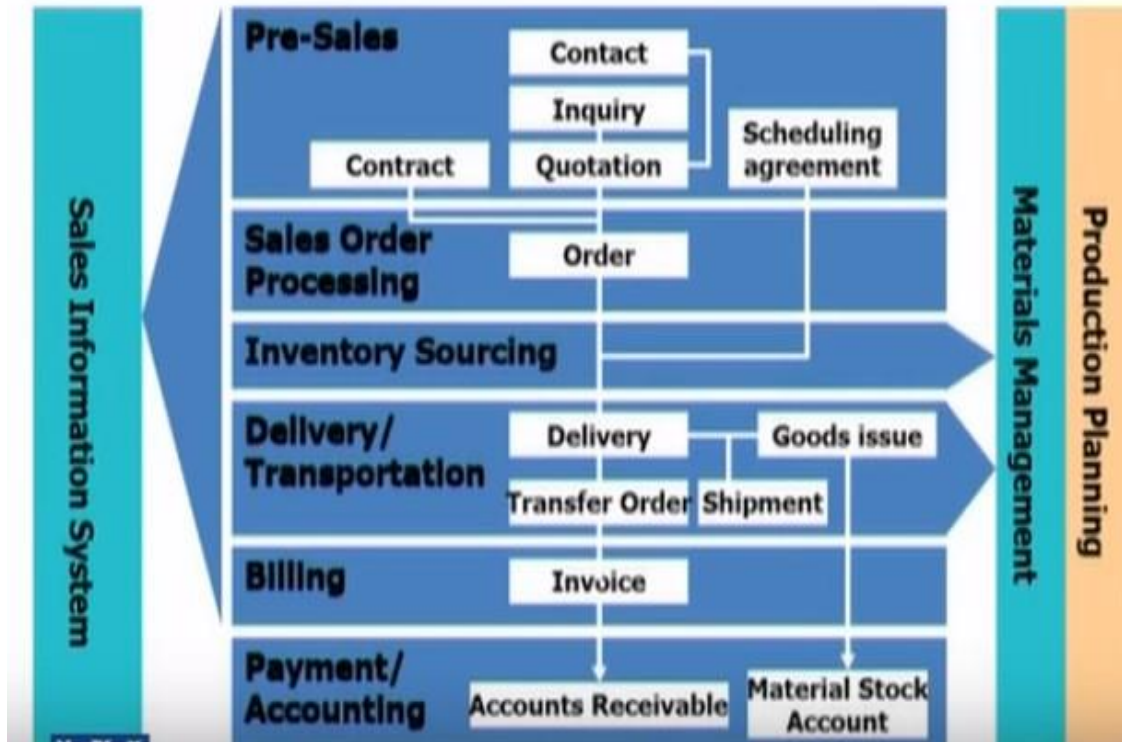
Data can also be manipulated from an inquiry or quote.

Standard Order in SAP ERP

Task is completed by

1. Order entry screen in SAP ERP's Enterprise System
2. A unique number is assigned by the company to each customer
3. For most data entry fields, SAP ERP determines whether an entry is valid

Order Management Process



What is CRM?

- CRM stands for Customer Relationship Management.
- It is a software system that is used to effectively manage the
 - sales process,
 - track customer interactions,
 - store information about customers (including purchase history, revenue generated, up-selling and cross-selling opportunities, etc.), and
 - ultimately strengthen relations with customers.
- CRM, however, is not just about software systems. It is a philosophy for interacting with your clients to make sure that they are happy and that they keep coming back to use your products and services.

CRM Sales Management Functionality

A CRM system supports your sales team at every stage of the sales cycle, from leads to customer management. Below are a few use cases:

- View and manage account activity and communications
- Use reports to forecast sales, measure business activity, identify trends
- Qualify leads and track prospective customers
- Centralize customer data
- Access, update, and share information across teams and departments.

Core Activities of CRM

The core activities of CRM system may include:

- One-to-One Marketing
- Call Center Automation
- Sales Force Automation (SFA)
- Sales Campaign Management
- Contact Management Tool
- Sales Activity Manager

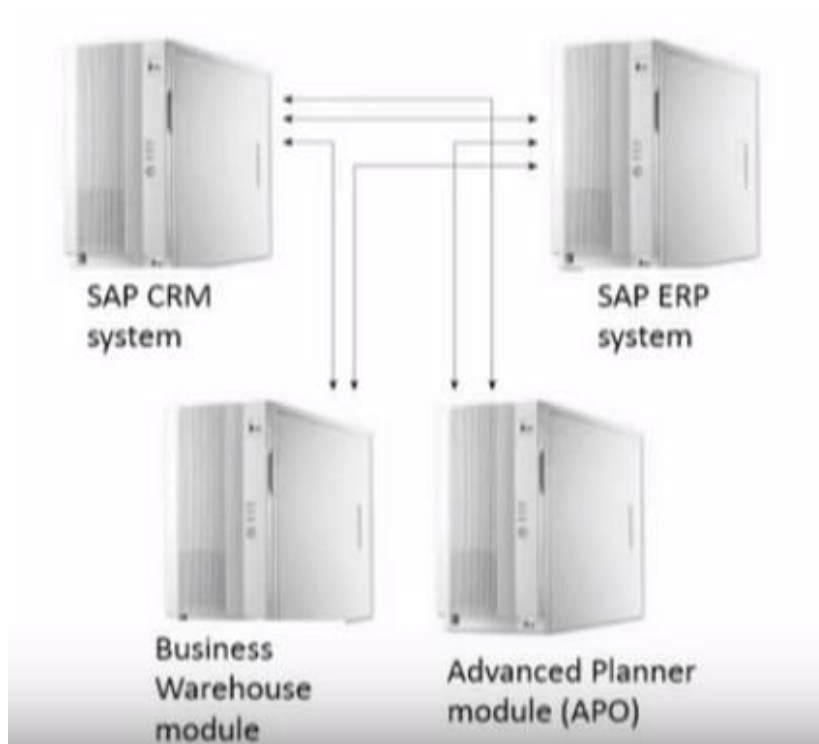
Key components of a successful CRM



SAP CRM System

- **SAP's Business Warehouse:** It is a system for reporting and analyzing transactional data relationship with the customer.
- **Advanced Planner and Optimizer (APO):** It is a system that supports efficient planning of the supply chain SAP's view of CRM to provide a set of tools to manage the three basic task areas:
 1. Marketing,
 2. Sales, and
 3. Service.

SAP CRM System



Marketing Automation

- **Marketing Automation**

It can be described as **event-based trigger** marketing that is used to launch messaging and offer presentations to customers at particular points in time. CRM helps not only trigger the communications, but also measure the results.

Double Opt-in

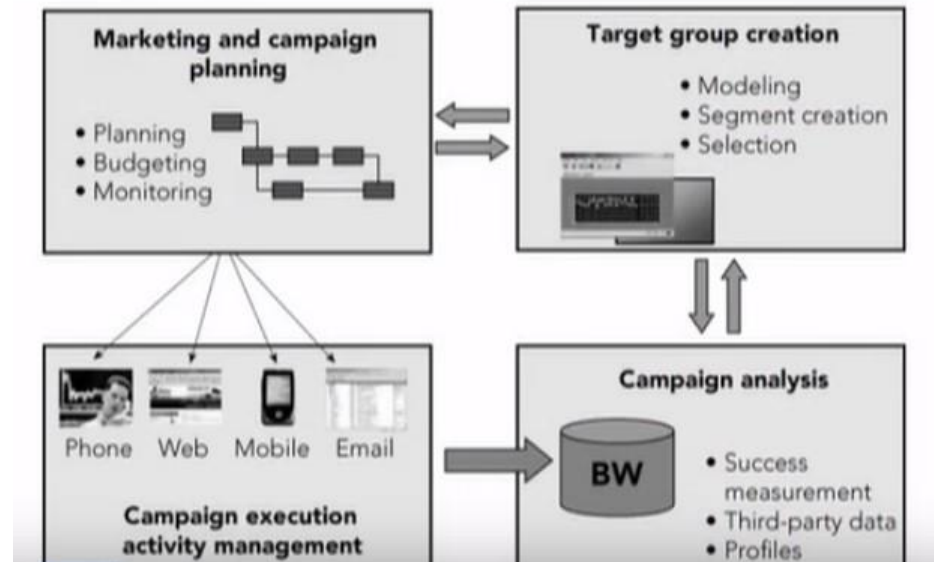
The **double opt-in** process implies that a subscriber fills out your signup form first and then confirms subscription to your list. In other words, the double opt-in works as follows:

1. A prospect signs up to receive emails via your website, at a tradeshow etc.
2. You send a confirmation email to him/her asking them to click on a link that confirms that they have opted to subscribe to your messages.
3. You get the double opt-in permission from prospects.
 - This eliminates the chance of abuse where somebody submits somebody else's email address without their knowledge or against their will.
 - Note that every communication you send must have the option for the contact to unsubscribe at any time.

SAP CRM System

Four phases of the cultivation of the customer relationship:

1. Prospecting,
2. Acquiring,
3. Servicing,
4. Retaining



Key Differences between CRM and SFA

- From a software perspective, SFA (Sales Force Automation) often refers to a primary component of CRM.
 - SFA is typically used by sales people and managers and covers functions such as
 - contact management,
 - account management,
 - activity management and
 - opportunity management.
- Sometimes SFA systems also include capabilities such as
 - sales order processing,
 - Partner Relationship Management (PRM), etc.

Key Differences between CRM and SFA

SFA

- Providing actionable goals and targets for each sales team member
- Recording cold calls and lead generation activities
- Storing contact information
- Scheduling and tracking appointments
- Managing pipelines, opportunities, and workflows

CRM

- Customer profiling including purchasing history, behavior data, demographics and more.
- Customer communications tracking
- Storing account data and e-commerce transactions.
- Tracking responses to sales and marketing campaigns
- Monitoring service and support interactions

Benefits of CRM System

- A CRM system can help you
 - recognize prospective customers,
 - learn more about current customers,
 - understand their preferences,
 - frequently anticipate their needs and respond to their requests quickly and effectively.
- In other words, with a CRM system, you can **track, organize, and consolidate your interactions with clients.**

Note:

When should I start tracking my customers' activities?

This is best to start doing before they actually become your customers. This will help you maximize the ROI and retain each customer.

Benefits of CRM for Businesses

With a CRM system, your organization will be able to improve productivity, reach out to more prospects and close more sales. CRM can help you:

- Raise customer satisfaction
- Increase customer retention
- Reduce marketing expenses
- Anticipate customer needs and preferences
- Increase operating efficiencies
- Improve targeted marketing efforts of customers and prospects
- Provide quicker service to customers.

Note:

Is CRM required for small business with limited customers?

CRM software can benefit small businesses by consolidating customer data into a single system. As your business grows, keeping a record of all transactions can become very challenging. CRM will allow you to manage customer interactions more efficiently, so you have more time to focus on your service or product.

E-mail Management through CRM

- CRM software may provide **workflow-enabled email processing capabilities**.
 - It can retrieve emails sent from your customers,
 - automatically route them to appropriate users based on workflow rules,
 - send auto replies back to your customers,
 - automatically associate emails with incidents and customers,
 - manage multiple attachments in emails, etc.
- Examples of Automated Messages:
 - Automated welcome messages to new customers, personalized with their most recent transactions
 - Thank-you messages to recognize your customers' activities
 - Lapsed messages with an irresistible offer to re-activate former highly valuable customers
 - Cross-sell messages based on previous purchases
 - Run marketing campaigns that rely on repeated contact or several 'touches' with potential customers (ideal for high-value products with a long sales cycle).

Difference between on-premise and cloud-based CRM solutions

- **What is “cloud”?**

The 'cloud' is a popular word for using the Internet to access data, stored on remote servers, i.e. like somewhere in the cloud. The cloud makes it convenient to get at information and services from anywhere.

- **Difference between on-premise and cloud-based CRM solutions.**

- As the name suggests, **on-premise CRM software** is run on computers within the premises of an organization.
- In this case all the data and information is stored inside the premises of the company, too.
- In recent years there has been a steady growth in the popularity of web-based solutions.
- Cloud-based software implies that the software and all relevant data, is accessible through the Internet and is displayed in a web browser.
- According to Gartner, 35% of all CRM implementations today use SaaS, growing to over 50% by 2020.

Cloud Based CRM



Salesforce Demo: <https://www.youtube.com/watch?v=A7AEc-B2PKQ> (3 mins)

Major CRM Vendors

Figure 1: Magic Quadrant for the CRM Customer Engagement Center



Source: Gartner (June 2021)

Cloud Based ERP Solutions

- Choosing between cloud-based (SaaS) and on-premises ERP solutions is a critical decision for organizations.
- Each approach has its own set of advantages and disadvantages.
- Additionally, some organizations opt for hybrid solutions that combine elements of both.

Cloud-Based (SaaS) ERP:

- **Deployment:** Cloud ERP is hosted and maintained by a third-party provider, and users access it through the internet.
- **Cost Structure:** Typically, cloud ERP is subscription-based, with ongoing monthly or annual fees.
- **Accessibility:** Users can access the system from anywhere with an internet connection, facilitating remote work and collaboration.
- **Updates and Maintenance:** The provider is responsible for updates, maintenance, and security patches.
- **Scalability:** Cloud ERP solutions are often more scalable, allowing organizations to adjust resources as needed.

vs. On-Premises ERP:

- **Deployment:** On-premises ERP is installed and maintained on an organization's own servers and infrastructure.
- **Cost Structure:** Initial costs include software licenses, hardware, and implementation. Ongoing costs involve maintenance and support.
- **Control:** Organizations have full control over their ERP environment, which may be essential for compliance and customization.
- **Accessibility:** Access is typically limited to the organization's premises, which can hinder remote work.
- **Updates and Maintenance:** Organizations are responsible for managing updates, maintenance, and security measures.
- **Scalability:** Scalability may be more complex and require additional investments in hardware and resources.

Pros and Cons

Cloud-Based (SaaS) ERP:

- *Pros:* Lower initial costs, accessible, rapid deployment, scalability, automatic updates – think of the flexibility of Amazon Prime Video.
- *Cons:* Ongoing costs, potential security concerns, limited customization, internet reliance – similar to the limitations of streaming services.

On-Premises ERP:

- *Pros:* Complete control, high customization, more control over data security, predictable costs – akin to owning a vast library of DVDs.
- *Cons:* High initial costs, slower deployment, maintenance responsibility, limited remote access – like the challenges of managing a physical media collection.

Hybrid Solutions

- **Flexibility:** Choose where data resides, similar to how Amazon combines online and physical retail strategies.
- **Cost-Effectiveness:** Utilize cloud resources for specific functions.
- **Integration:** Blend on-premises and cloud systems.
- **Scalability:** Adapt to changing needs.

How to implement ERP

1. Define Objectives and Goals:

- Conduct a thorough needs analysis to understand the specific challenges and opportunities within your organization.
- Establish clear, measurable goals for the ERP implementation, aligning with overall business objectives.

2. Choose the Right ERP System:

- Perform a comprehensive assessment of various ERP solutions to identify one that best fits your business requirements.
- Engage with multiple vendors, request demonstrations, and conduct pilot tests to evaluate the functionality and user-friendliness of the systems.

3. Form an Implementation Team:

- Assemble a cross-functional team comprising members from various departments like IT, finance, HR, and operations.
- Appoint a skilled project manager to lead the implementation, ensuring they have the authority and resources to make key decisions.

How to implement ERP

4. **Develop a Detailed Plan:**

- Outline a detailed implementation timeline, including major milestones and deadlines.
- Allocate necessary resources, including personnel, technology, and budget, for each phase of the project.
- Include contingency plans to address potential delays or challenges.

5. **Data Migration and Cleansing:**

- Conduct a comprehensive review of existing data to identify and cleanse any inaccuracies or redundancies.
- Develop a detailed data mapping plan to align existing data structures with the new ERP system.
- Choose efficient data migration tools and techniques to ensure a smooth transition.

6. **Configuration and Customization:**

- Customize and configure the ERP system to align with specific business processes and workflows.
- Engage key users in the customization process to ensure the system meets their needs and expectations.

How to implement ERP

7. Training and Change Management:

- Develop a comprehensive training program for all users, tailored to different roles and departments.
- Implement a change management strategy to address employee concerns and resistance, ensuring clear communication about the benefits and changes brought by the ERP system.

8. Pilot Testing:

- Conduct a pilot phase with a select group of users to test the functionality and performance of the ERP system in a controlled environment.
- Collect and analyze feedback to make necessary adjustments before full deployment.

9. Full Deployment:

- Implement the ERP system across the entire organization, ensuring all necessary integrations and data migrations are complete.
- Provide ongoing support and resources to address any issues during the transition.

How to implement ERP

10. **Monitor and Optimize:**

- Establish a system for ongoing support and maintenance, addressing any technical issues promptly.
- Continuously monitor the system's performance and user feedback to identify areas for improvement.

11. **Evaluate and Refine:**

- Conduct regular evaluations of the ERP system to assess its impact on business processes and objectives.
- Make iterative refinements and updates to ensure the system remains aligned with evolving business needs and technology advancements.

ERPNext: Cutting-edge Enterprise Management Tool

- <https://aws.amazon.com/marketplace/pp/prodview-evtwbtmdkl3ya>

In-Class: ERP Lab (BlueFin eFoils)

BlueFin eFoils, Inc. is a fictional maker of electric hydrofoil boards. Demand for its BF-100 kit has jumped and stock is tight.

Get `In-Class_ERP-Lab_BlueFin.ipynb` from the In-Class: ERP Lab assignment on Canvas and open it in Google Colab. Group work, one notebook per project team.

Four required parts:

- Part 1: map Order-to-Cash and Procure-to-Pay. Name the modules and master data, one risk and one control for each.

- Part 2: Order-to-Cash. Allocate stock first come, first served, then invoice.

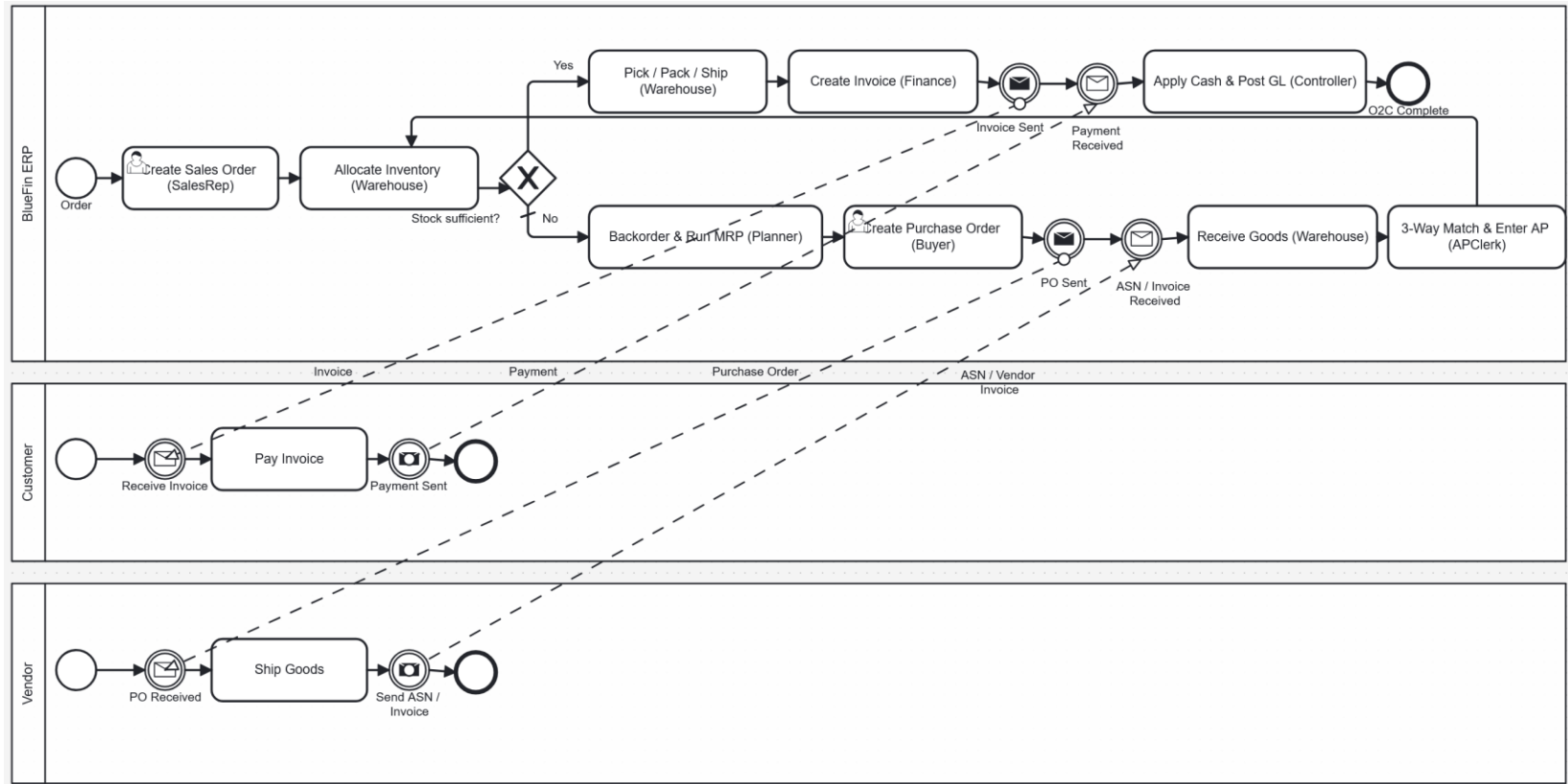
- Part 3: clean and de-duplicate the vendor master data.

- Part 4: roles and segregation of duties. Write the access check and the conflict check.

Optional: reorders and receiving, a one-level MRP, and a KPI plot.

Each code part has a CHECK cell that prints PASS or says what to fix. Save a copy in Drive first. Restart and run all, then upload the .ipynb. Due tonight, 11:59 PM.

ERP System BPMN Model



HW #5 Salesforce Assignment

- This is an individual assignment.
- Complete the “Developer Beginner” trail on Trailhead (required: 6,500 points):
- https://trailhead.salesforce.com/trail/force_com_dev_beginner
- Submit a Word document with screenshots from your Trailhead account showing module progress, badges and points earned.
- **Due: section 49 Sunday, September 27, 11:59 PM. Section 03 Sunday, October 4, 11:59 PM. Canvas has the date.**

References

- <https://www.oracle.com/webfolder/assets/ebook/complete-guide-to-modern-erp/index.html>
- <http://www.doungengelbart.org/pubs/augment-3906.html>
- <http://nexus.awakentech.com:8080/at/awaken1.nsf/UNIDs/CFB70C1957A686E98825654000699E1B?OpenDocument>
- <https://www.tableau.com/sites/default/files/media/designing-great-visualizations.pdf>
- <http://proquest.safaribooksonline.com/book/operations/9788131760994>
- <https://www.safaribooksonline.com/library/view/identity-management-a/9781583470930/>
- <https://www.safaribooksonline.com/library/view/core-concepts-of/9781118022306/>
- <https://www.mulesoft.com/resources/esb/erp-integration-application-architecture>

SAN JOSÉ STATE UNIVERSITY *powering* SILICON VALLEY